

HCSM-00

Foundations, Notation, and Mapping

The Hodge Complex Standard Model

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Abstract

The Hodge Complex Standard Model (HCSM) derives the gauge structure, matter content, particle spectrum, mixing matrices, spacetime dimension, graviton, and Newton constant of the Standard Model from a single input: the spacetime dimension $d = 4$. The dimension itself is derived from the photon's two helicity states and independently from the equal entropy spacing of the torus Laplacian. The substrate is the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, on which the Hodge complex of differential forms carries the Hodge–Dirac operator $D = d + d^*$ with $D^2 = \Delta$ exactly on each form degree. The 14 modes at the self-paired eigenvalue $\lambda = 4$ are identified with one Standard Model generation. This paper establishes the framework's foundational objects, states the notation, and gives a systematic mapping from the prior WIN-era papers to the HCSM paper stack. For each prior paper, we assign one of four verdicts *retained*, *reframed*, *sharpened*, or *retired* and we state which HCSM paper supersedes it. All theorems cited here are HCSM native: they use only the 8×8 torus, the Hodge complex, the D_4 action, the discrete Laplacian, the winding generator, and the zero state. No imported machinery (path integrals over continuum fields, SYK kernels, Ryu–Takayanagi strip integrals, Kaluza–Klein towers) is used in any theorem stated as HCSM-native.

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1 Introduction

1.1 What HCSM is

The Hodge Complex Standard Model (HCSM) is a structural derivation of the Standard Model of particle physics from a single discrete substrate: the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. The framework's central claim is that the Standard Model is not a fundamental description of nature, but the residue left behind when a small, discrete, information-bearing lattice cancels itself against its own mirror.

The one line summary, which recurs throughout the HCSM paper stack, is:

Physics is what the machine cannot cancel. The Hodge complex is the machine.

HCSM does not ask what fields exist, or what symmetry groups are broken. It asks a single question: what substrate could produce exactly the Standard Model, and nothing else? The framework's answer is a discrete lattice whose mirror cancellation leaves exactly the observed particle content.

1.2 The single input: $d = 4$

The whole structure of HCSM traces back to a single input: the spacetime dimension $d = 4$. This input is itself derived from the photon's two helicity states, and the substrate size $L = 8$ is selected independently by the equal-entropy-spacing theorem on the torus Laplacian. The two derivations agree, and together they force the substrate and the emergent dimension.

1.3 The three part structure of HCSM

The HCSM paper stack is organized in three parts:

1. **The substrate and its spectral content.** The 8×8 torus, the Hodge complex, the 14-mode multiplet at $\lambda = 4$, and the D_4 representation theory that determines the gauge algebra.
2. **The machine's operation.** The processor T and its F/B split, the winding generator and the class partition, the loop operator, and the residual charge, mass, and mixing structure.
3. **The outputs.** Gravity, the Newton constant, the dark energy density, and the falsifiable predictions.

This foundation paper (HCSM-00) states the substrate, the Hodge complex, the spectral midpoint, and the processor, then gives the mapping from the prior WIN-era paper stack to the HCSM paper stack. Everything downstream is stated in the individual HCSM papers.

1.4 Scope and honesty

Every theorem stated as HCSM native in this paper and in the downstream HCSM stack uses only the following primitives:

- the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$,
- the Hodge complex $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$ of differential forms,
- the Hodge-Dirac operator $D = d + d^*$,
- the D_4 point-group action on Λ ,
- the discrete 5-point Laplacian Δ ,
- the winding generator $H_{\text{wind}} = i \sin(\pi M/8)$,
- the zero-state vector s from the unique nullspace of a 14×19 vacuum-native measurement matrix,
- the charge functional $q_3 = F(n_1, \sigma)$,
- the $N/4$ labels from the δ function,

- the processor $T = P_{\text{mesh}} + P_{\tau}$ and its associated class partition.

No imported machinery appears in any HCSM-native theorem. Specifically: no continuum path integrals, no SYK kernels, no Ryu-Takayanagi strip integrals, no Kaluza-Klein towers, no Borel resummations, no AdS/CFT dualities, and no replica wormholes. Where prior WIN era papers used such machinery, this document states explicitly that the layer is retired or pending, and names the HCSM native theorem that replaces it or the open problem that remains.

1.5 Organization of this paper

Section 2 states the substrate and the two independent anchors that select it, then derives the third anchor as a consequence. Section 3 states the Hodge complex and the Hodge-Dirac operator. Section 4 states the spectral midpoint and the 14-mode multiplet. Section 5 states the processor T , the class partition, and the F/B split. Section 6 gives a compact overview of the output structures. Section 7 gives the paper-by-paper mapping from the WIN-era stack to the HCSM stack. Section 8 gives the notation translation table. Section 9 states what is retired. Section 10 lists the paper stack and the current status of each paper. Section 11 gives reproducibility notes.

2 The substrate

2.1 The periodic 8×8 torus

The substrate of HCSM is the periodic square lattice

$$\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8, \quad (1)$$

consisting of $N = 64$ sites arranged in an 8-by-8 grid with periodic boundary conditions. Site (x, y) is identified with $(x + 8, y)$ and with $(x, y + 8)$; equivalently, the coordinates x and y are each defined modulo 8.

The lattice is not a discretization of an underlying continuum. It is not a finite approximation to a smooth manifold. It is not a mesh on top of a smooth space. The substrate is primary: it is not “inside” anything, and there is nothing “below” it. Its 64 sites are a structural count, like the number of vertices of a cube, not a spatial extent.

Topologically, the periodic lattice is equivalent to the torus $T^2 = S^1 \times S^1$.

2.2 The discrete 5-point Laplacian

Each site (x, y) has four nearest neighbors: $(x \pm 1, y)$ and $(x, y \pm 1)$, with all coordinates modulo 8. The natural differential operator on the lattice is the discrete 5-point Laplacian

$$(\Delta f)(x, y) = 4f(x, y) - f(x + 1, y) - f(x - 1, y) - f(x, y + 1) - f(x, y - 1). \quad (2)$$

The normalization 4 in front of $f(x, y)$ is chosen so that the constant function $f(x, y) = 1$ is annihilated by the Laplacian: $\Delta 1 = 0$.

The eigenvalues of Δ on plane waves $\phi_{n_1, n_2}(x, y) = e^{2\pi i(n_1 x + n_2 y)/8}$ are

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right), \quad n_1, n_2 \in \{0, 1, \dots, 7\}. \quad (3)$$

The spectrum is mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$, with 13 distinct eigenvalues.

2.3 The two independent anchors and the derived third anchor

The size $L = 8$ is not put in by hand. It is determined by two independent anchors. A third anchor follows as a theorem.

Anchor 1 (input): photon helicity.

A massless spin-1 particle has exactly two helicity states. Each is described by a complex amplitude with two real components. Two states $\times$ two components = 4 real numbers, spanning a four dimensional phase space whose Wick rotation produces four dimensional Lorentzian spacetime. Hence $d = 4$.

Anchor 2 (input): equal entropy spacing.

For an $L \times L$ torus, each mode has a Shannon entropy in bits. The modes fall into entropy classes. Among all integers $4 \leq L \leq 10^4$, the only L for which the torus Laplacian has exactly three equally spaced entropy classes is $L = 8$. The three classes are 6.0, 5.5, 5.0 bits, with spacing 0.5 bits. Hence $L = 8$.

Anchor 3 (derived): topology of the kernel.

Given the substrate $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ selected by Anchors 1 and 2, the kernel of the Hodge-Dirac operator D on Λ has dimension $b_0 + b_1 + b_2 = 1 + 2 + 1 = 4$, where (b_0, b_1, b_2) are the Betti numbers of T^2 . This is stated and proved in Section 3 as Theorem 3.5. It is *not* an independent input; it is a consequence of the substrate selected by Anchors 1 and 2.

The logical dependency graph is therefore:

$$\text{Anchor 1} \longrightarrow d = 4, \quad \text{Anchor 2} \longrightarrow L = 8, \quad \{d = 4, L = 8\} \longrightarrow N = d \cdot 2^d = 64 \longrightarrow \text{Anchor 3.} \quad (4)$$

The consistency of $d = 4$ with $L = 8$ follows from the framework relation $N = d \cdot 2^d$, which gives $N = 64$ at $d = 4$ and $L = \sqrt{N} = 8$. Deriving the relation $N = d \cdot 2^d$ from first principles is the subject of HCSM-01 and is currently listed as a pending item (Section 10). At the level of HCSM-00, we take $N = d \cdot 2^d$ as a framework relation and note that it is consistent with both anchors.

2.4 The 14 mode count at the self paired eigenvalue

The torus has exactly 14 modes at the self paired eigenvalue $\lambda = 4$:

$$\text{multiplicity}_{\lambda=4} = 2(L - 1) = 2(8 - 1) = 14. \quad (5)$$

This matches one Standard Model generation. For any other L , the count is wrong. The torus is therefore not merely preferred by the entropy anchor; it is unique.

3 The Hodge complex

3.1 Differential forms on the torus

On the two-dimensional torus Λ , there are three types of differential forms:

- Ω^0 : 0-forms (functions on the 64 sites), dimension 64, bosonic;
- Ω^1 : 1-forms (fields on the 128 directed edges), dimension 128, fermionic;
- Ω^2 : 2-forms (fields on the 64 plaquettes), dimension 64, bosonic.

The total dimension is $64 + 128 + 64 = 256$.

3.2 Exterior derivative and codifferential

For a 0-form f and a 1-form $\omega = (\omega_1, \omega_2)$, the exterior derivatives are

$$(d_0 f)(x, \mu) = f(x + e_\mu) - f(x), \quad (6)$$

$$(d_1 \omega)(x) = \omega_2(x + e_1) - \omega_2(x) - \omega_1(x + e_2) + \omega_1(x). \quad (7)$$

The codifferential $d^* = d^\dagger$ is the adjoint with respect to the standard inner products. They satisfy exact nilpotency: $d_1 d_0 = 0$.

3.3 The Hodge-Dirac operator

[Hodge-Dirac operator] The Hodge-Dirac operator is

$$D = d + d^*, \quad (8)$$

acting on $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$.

[Hodge-Dirac square] On the discrete torus Λ , the identity

$$D^2|_{\Omega^k} = \Delta, \quad k = 0, 1, 2, \quad (9)$$

holds exactly on each form degree, where Δ is the discrete 5-point Laplacian of Eq. (2).

The identity $D^2 = \Delta$ is exact, not approximate. Unlike the spinorial Dirac operator on the same lattice, which suffers from fermion doubling, the Hodge-Dirac square is the ordinary scalar Laplacian.

3.4 The $\mathbb{Z}_2$ grading

Even forms ($\Omega^0 \oplus \Omega^2$) are bosonic; odd forms (Ω^1) are fermionic. The Hodge-Dirac operator is odd with respect to this grading: it is the supercharge. This $\mathbb{Z}_2$ grading is the origin of the spin statistics connection in HCSM (see HCSM-42 for the full derivation).

3.5 The kernel and the emergence of four dimensions

[Kernel dimension] By the discrete Hodge theorem, the kernel of D is isomorphic to the de Rham cohomology of the torus:

$$\ker D \cong \bigoplus_{k=0}^2 H^k(T^2; \mathbb{R}), \quad (10)$$

with dimension $\dim \ker D = b_0 + b_1 + b_2 = 1 + 2 + 1 = 4$.

The four harmonic forms of Theorem 3.5 are the four spacetime directions. This is the derivation of Anchor 3 from the substrate selected by Anchors 1 and 2; it is a theorem of the Hodge complex, not an independent input.

The canonical metric on $\ker D$ is constructed from the intersection form $Q(\alpha, \beta) = \int \alpha \wedge \beta$ on $H^0 \oplus H^2$ and the Hodge inner product $\langle \alpha, \beta \rangle = \int \alpha \wedge \star \beta$ on H^1 . The resulting metric has signature $(3, 1)$ with no free coefficients. The timelike direction is the self-dual 0-form/2-form combination $(v_0 + v_2)/\sqrt{2}$. See HCSM-03 for the full derivation.

4 The spectral midpoint and the 14-mode multiplet

4.1 The self paired eigenvalue

Solving $\lambda = 8 - \lambda$ in the mirror symmetry of the Laplacian spectrum gives the self-paired midpoint $\lambda = 4$. This eigenvalue has multiplicity 14 and cannot cancel with any mirror partner. The 14 modes at $\lambda = 4$ are the residue that survives.

4.2 The three D_4 orbits

The 14 modes at $\lambda = 4$ decompose under the D_4 point group of the square into three orbits:

$$\mathcal{O}_0 = \{(0, 4), (4, 0)\}, \quad \text{size 2, support 64,} \quad (11)$$

$$\mathcal{O}_1 = \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \quad \text{size 8, support 48,} \quad (12)$$

$$\mathcal{O}_2 = \{(2, 2), (2, 6), (6, 2), (6, 6)\}, \quad \text{size 4, support 32.} \quad (13)$$

The support classes are $\{64, 48, 32\} = \{N, V, N/2\}$ with $V = (d-1) \cdot 2^d = 48$. The inverse participation ratios (IPR) are exactly $2/128, 3/128, 4/128$ on the three orbits. The Shannon entropies are exactly 6.0, 5.5, 5.0 bits, with equal spacing 0.5 bits.

4.3 The Hodge lift to the 56-dimensional sector

The 14-fold degeneracy at $\lambda = 4$ lifts to the full Hodge complex:

$$14\Omega^0 + 28\Omega^1 + 14\Omega^2 = 56. \quad (14)$$

This 56-dimensional sector is the arena of the physical world. Its D_4 content is

$$\Omega^0|_{\lambda=4} \oplus \Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4} = 9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E. \quad (15)$$

5 The processor and the F/B split

5.1 The winding generator

The processor acts on the 14-mode multiplet at $\lambda = 4$. Its first move is the winding generator

$$H_{\text{wind}} = i \sin\left(\frac{\pi}{8}M\right), \quad M_{ij} = n_1^{(i)}n_2^{(j)} - n_1^{(j)}n_2^{(i)}, \quad (16)$$

where $(n_1^{(i)}, n_2^{(i)}) \in \mathbb{Z}_8 \times \mathbb{Z}_8$ are the momentum labels of mode i . Because $n_1, n_2 \in \mathbb{Z}_8$, the argument $(\pi/8)M_{ij}$ is an integer multiple of $\pi/4$, so the entries of $\sin((\pi/8)M)$ lie in $\{0, \pm 1/\sqrt{2}, \pm 1\}$ exactly. The matrix H_{wind} is therefore Hermitian with entries in $\mathbb{Q}(\sqrt{2})$.

The characteristic polynomial of H_{wind} restricted to the 14-mode subspace is

$$p(\lambda) = \det(\lambda I - H_{\text{wind}}) = \lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32) = \lambda^{14} - 40\lambda^{12} + 256\lambda^{10}. \quad (17)$$

The spectrum is

$$\text{spec}(H_{\text{wind}}) = \{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}, \quad (18)$$

with multiplicities 1, 1, 10, 1, 1. The characteristic polynomial is derived from the traces $\text{tr}(H_{\text{wind}}^2) = 80$ and $\text{tr}(H_{\text{wind}}^4) = 2176$ via Newton's identities, and is verified at machine precision in the companion verification script. See HCSM-14 for the full derivation.

5.2 The three winding classes

The 14 modes partition into three classes by frequency:

$$\text{Class } A : \{(0, 4), (4, 0), (6, 6)\}, \quad \omega = 2\sqrt{2}, \quad (19)$$

$$\text{Class } B : \{(1, 3), (1, 5), (3, 1), (3, 7), (7, 3), (7, 5), (5, 1)\}, \quad \omega = 6\sqrt{2}, \quad (20)$$

$$\text{Class } C : \{(5, 7), (2, 2), (2, 6), (6, 2)\}, \quad \omega = 0. \quad (21)$$

The class projectors P_A, P_B, P_C are diagonal in the mode basis, orthogonal, idempotent, and sum to the identity.

5.3 The processor T

The processor T is the rank-10 orthogonal projection

$$T = P_{\text{mesh}} + P_{\tau} = I - P_{\text{sub}} + P_{\tau}, \quad (22)$$

where P_{mesh} projects onto the 9 mesh resonances ($\delta \neq 0$) and P_{τ} projects onto the unique mode with zero-state amplitude $s = +1$. The output is the fermion/boson split

$$F = \{\delta \neq 0\} \cup \{s = +1\}, \quad |F| = 10, \quad (23)$$

$$B = \{\delta = 0\} \setminus \{s = +1\}, \quad |B| = 4. \quad (24)$$

The 10 modes of F are the Standard Model fermions; the 4 modes of B are the Standard Model bosons.

5.4 The F/B -class correspondence

The class partition $\{A, B, C\}$ and the processor's F/B split are related by the exact correspondence

$$A \cup B = \text{image}(T) = F, \quad C = \ker(T) = B. \quad (25)$$

Equivalently, the processor $T = I - P_C$ is exactly the complement of the class- C projector. This is stated and proved in HCSM-13. It is a consequence of the class partition and the definition of T , not an input.

5.5 The Landauer channel

The Hodge complex delivers the maximally mixed state on the 56-dimensional sector, with entropy $S_{\text{in}} = \log 56$ nats. The processor reduces this to $S_{\text{out}} = \log 10 + \log 4 = \log 40$. The entropy reduction

$$\Delta S = \log 56 - \log 40 = \log(7/5) = 0.33647 \dots \text{ nats} \quad (26)$$

saturates the deterministic Landauer bound. The processor is a Maxwell demon operating at the exact deterministic limit.

5.6 The zero state

A 14×19 measurement matrix M of vacuum native observables has rank 13 and nullity 1. Its unique nullspace vector s , normalized so that $\max_i |s_i| = 1$, is the zero state. In the standard mode order

$$(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t), \quad (27)$$

the zero-state vector is

$$s = \begin{pmatrix} -0.105198199128, & +0.105198199128, & +0.937726561300, \\ -0.192806690737, & -0.937726561300, & +0.658283218429, \\ -1.000000000000, & -0.658283218429, & +1.000000000000, \\ +0.192806690737, & 0.000000000000, & -0.329141609215, \\ +0.329141609215, & 0.000000000000 \end{pmatrix}. \quad (28)$$

The vector is odd under the diagonal reflection $(n_1, n_2) \leftrightarrow (n_2, n_1)$. The two fixed points $(2, 2)$ and $(6, 6)$ carry $s = 0$ exactly. These two modes are the observer pair $\{W, t\}$, and their properties form the subject of HCSM-26.

6 Overview of the output structures

This section gives a compact overview of the output structures that the HCSM paper stack develops in full. Each item below is stated with a forward reference to the HCSM paper that derives it.

6.1 Charge

The charge theorem states that the electric charge in units of $e/3$ is a bijection

$$q_3 = F(n_1, \sigma), \quad n_1 \in \mathbb{Z}_8, \quad \sigma = \text{sign}(s) \in \{-1, 0, +1\}, \quad (29)$$

from the 14 realized pairs to the five Standard Model charge eigenvalues $\{-3, -1, 0, +2, +3\}$. The three σ -lines carry charges 0, +5, -8, summing to $-1e$. The associated charge density $\rho(x, y)$ on the torus is exactly 4×4 periodic. See HCSM-24 and HCSM-25.

6.2 Mass

The mass map is

$$m_i = A_{\text{mass}} \exp(-B_{\text{mass}} \cdot (N/4)_i), \quad (30)$$

with $A_{\text{mass}} = (2d - 1)N/2 = 224$ and $B_{\text{mass}} = (d - 1)kL^2/(2N^2) = 0.541197$. The map reproduces the 12 charged fermion masses at RMS 0.0449 dex ($\sim 11\%$ accuracy) with zero free parameters. The $N/4$ labels are derived from the mode coordinates by the δ function. See HCSM-23 and HCSM-33.

6.3 Flavor

The CKM matrix is derived from the D_4 breaking pattern and the zero state, with zero free parameters. Eight of nine magnitudes are reproduced within 4%; the Jarlskog invariant J is reproduced to 0.05%. The PMNS matrix is derived from the neutrino $N/4$ spectrum and is reproduced at RMS 0.035 across all nine elements. See HCSM-34, HCSM-35, HCSM-36, HCSM-37.

6.4 Gravity

The graviton is the self-paired $B_1 \oplus B_2$ curvature mode of $\text{Sym}^2(\Omega^1|_{\lambda=4})$, consisting of $54B_1 \oplus 54B_2 = 108$ modes. It is exactly massless at all orders by D_4 Schur orthogonality. The Newton constant is

$$G_N = \frac{25}{8\pi M_P^2} = \frac{(d+1)^2}{8\pi M_P^2}, \quad (31)$$

with the factor $25 = (d+1)^2$ from the bilinear trace of the hidden B_1 isotypic sector and $8\pi = 2\pi \dim \ker D$ from the Faddeev–Popov measure on the discrete torus. See HCSM-45 through HCSM-50.

6.5 Cosmology

The kL crossover fixes the warp factor self-consistently at $kL = 38.440831$, matching $\ln(M_P/v_{\text{EW}}) = 38.442527$ to 0.004%. The dark energy density is

$$\rho_{\text{DE}} = \frac{|F|}{|B|} \cdot m_\nu^4 = \frac{d+1}{2} \cdot m_\nu^4, \quad (32)$$

with $|F| = 10$, $|B| = 4$, and $m_\nu = 1.7749$ meV. This gives $\rho_{\text{DE}} = 2.4809 \times 10^{-47}$ GeV⁴, matching the observed value to 0.76%. The equation of state is $w = -1$ exactly from topological Casimir. See HCSM-51 through HCSM-54.

7 Mapping from the WIN-era paper stack

This section gives a paper by paper mapping from the WIN era paper stack to the HCSM paper stack. Each row states the WIN era paper, the corresponding HCSM paper, and one of four verdicts:

Retained.

The paper’s content is HCSM-native as stated and carries forward to HCSM without change.

Reframed.

The paper’s mathematical content is HCSM-native, but the notation, terminology, or framing changes.

Sharpened.

The paper’s content is correct, but a new HCSM theorem adds exactness or closes a previously open item in the paper’s own domain.

Retired.

The paper used imported non-HCSM machinery that has been replaced by an HCSM-native theorem. The retired layer is not used by any HCSM theorem.

Table 1: Mapping from the WIN-era paper stack to the HCSM paper stack.

WIN-era paper	HCSM paper	Verdict	Notes
WIN Paradigm Explained, Version 3	HCSM-00	Reframed	This document
Derivation of Spacetime Dimension from the Photon’s Two Helicity States	HCSM-01	Retained	$d = 4$ anchor
Derivation of the 8×8 Torus Substrate	HCSM-01	Sharpened	$N = d \cdot 2^d$ pending
The Hodge Complex on the 8×8 Torus	HCSM-02	Retained	$D^2 = \Delta$
Lorentzian Signature from the Hodge Complex	HCSM-03	Retained	Hodge star on $\ker D$
Spacetime from the Kernel of the Hodge–Dirac Operator	HCSM-03	Retained	$\dim \ker D = 4$
The 14 Particles as the Residual of a Self-Canceling Vacuum Orbit Decomposition and Mean-Field Chemical Potential Shifts	HCSM-04, HCSM-10 HCSM-05	Retained Retained	Mirror-cancellation mechanism Support classes, IPR
Gauge Algebra and Hypercharge Structure	HCSM-06, HCSM-07	Retained	$\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$
Gauge Topology and Anomaly Closure	HCSM-09	Retained	Orientation flip
Sector-Coarse Correspondence	HCSM-06	Sharpened	Endomorphism-algebra derivation
5 + 9 Split of the 14-Mode Multiplet	HCSM-11	Retained	Substrate vs. mesh

WIN-era paper	HCSM paper	Verdict	Notes
Hodge–FDLM Interface	HCSM-12	Reframed	Landauer channel
Hodge–Dirac / processor T	HCSM-13	Sharpened	$T = I - P_C$
Winding generator spectrum	HCSM-14	Retained	Classes A, B, C
Gauge Symmetry of the Loop Operator and DNLS	HCSM-15	Retained	Commutant structure
Loop Operator on the 56-Dimensional Hodge Sector	HCSM-16	Retained	Block-diagonal extension
Numerical Cartan rotation	HCSM-17	Sharpened	Exact closed form (this session)
Dynamical Bridge: DNLS on the Torus	HCSM-18	Retained	Three-tier structure
DNLS Origin of the Mass Map	HCSM-19	Sharpened	$(d - 1)/2$ coefficient pending
WIN Motion Theorem, Version 2	HCSM-20	Sharpened	Loop eigenvalues, amplitudes
WIN Time Theorem, Version 2	HCSM-21	Sharpened	Loop pass duration pending
The Zero State of the 8×8 Torus	HCSM-22	Retained	Unique nullspace s
The δ Function on the 8×8 Torus	HCSM-23	Retained	14 $N/4$ values
Charge theorem	HCSM-24	Retained	$q_3 = F(n_1, \sigma)$
4×4 periodic charge density	HCSM-25	Retained	Row/column sums -12
Observer pair $\{W, t\}$	HCSM-26	Retained	This session
Klein orbit structure	HCSM-27	Retained	This session
Orbit-class reconstruction	HCSM-28	Retained	This session
Charge–entropy–observer triangle	HCSM-29	Retained	This session
Base-2 mass formula pattern	HCSM-30	Retained	Structural feature
Flavor Mechanics, Generations, and Mass-Scale Entropy	HCSM-31	Sharpened	Generation count pending
Shell Assignment Rule	HCSM-32	Retained	14/14 match
The Mass Map	HCSM-33	Sharpened	$(d - 1)/2$ coefficient pending
CKM Matrix	HCSM-34	Sharpened	Residual structure pending
V_{td} Structural Relations	HCSM-35	Sharpened	Non-perturbative derivation pending
PMNS Matrix	HCSM-36	Sharpened	0.23% overshoot pending
PMNS generation phase	HCSM-37	Sharpened	Phase formula pending
Hidden Sector Content	HCSM-38	Retained	5 B_1 , 6 A_1 , 31 singlets
Weinberg angle $\sin^2 \theta_W(M_Z) = 0.2312$	HCSM-39	Retained	Dual-role theorem
Higgs mechanism	HCSM-40	Sharpened	Higgs origin pending
Dark photon and kinetic mixing	HCSM-41	Retained	$\varepsilon = 2025/1697500$

WIN-era paper	HCSM paper	Verdict	Notes
Spin and chirality	HCSM-42	Sharpened	Spin structure pending
Strong CP	HCSM-43	Sharpened	Orientation $\mathbb{Z}_2$ pending
Proca sector	HCSM-44	Sharpened	$U(1)$ origin pending
The WIN Gravity Theorem	HCSM-45	Retained	Graviton as $54B_1 \oplus 54B_2$
Exact masslessness	HCSM-46	Retained	Schur orthogonality
Graviton self-coupling from D_4	HCSM-47	Retained	Counts $(0, 2, 0, 3)$, $(5, 12, 11)$
Faddeev–Popov G_N normalization	HCSM-48	Retained	$25 = (d + 1)^2$
Completion of the Newton Constant Sector	HCSM-49	Retained	$8\pi = 2\pi \dim \ker D$
Composite-Operator Effective Action	HCSM-50	Retained	Nine structural zeros
kL crossover (bare)	HCSM-51	Retained	$kL = 38.4$ exact
kL crossover (corrected)	HCSM-52	Sharpened	Needs DME removal
EFT coefficient in ρ_{DE}	HCSM-53	Sharpened	$c = 5/9$ derived (this session)
Dark energy density	HCSM-54	Sharpened	Needs HCSM-52
Dimensional Measurement Equilibrium	HCSM-55	Pending import removal	Ryu–Takayanagi strip integral
ENTROPIA-Core	HCSM-56	Pending import removal	EDF Ohmic bath
ENTROPIA-EFT	HCSM-57	Pending import removal	2D bubble, Proca ansatz
H_0 boost	HCSM-58	Pending	Observed only
SYK kernel and joint independence	HCSM-59	Retired	SYK removed from HCSM
ENTROPIX/MESA	-	Retired	Not referenced by any HCSM theorem
Kaluza–Klein graviton at 1.52 TeV	-	Retired	Category error (HCSM-49)
5D/6D/7D/8D tower	-	Retired	Not HCSM-native

8 Notation translation

The following table gives the notation translation from the WIN-era paper stack to the HCSM paper stack. Notation changes are cosmetic; the underlying objects are identical.

Table 2: Notation translation from WIN-era to HCSM.

WIN-era notation	HCSM notation
“FDLM/WIN processor”	“processor T ”
“FDLM substrate”	“substrate Λ ”
“Hodge-FDLM interface”	“Hodge–HCSM interface”
“SYK/QIN layer”	(retired)

WIN-era notation	HCSM notation
“DME stability coefficient γ ”	“ γ (pending HCSM-native derivation)”
“ENTROPIA-Core flow”	(retired)
“ENTROPIA-EFT”	(retired)
“ENTROPIX/MESA selection”	(retired)
“ $w_{\text{WIN}} = -1$ ”	“ $w = -1$ (topological Casimir)”
“WIN-native”	“HCSM-native”
“WIN Paradigm”	“Hodge Complex Standard Model (HCSM)”

9 Retired layers

Several layers from the WIN era paper stack used imported machinery that is not HCSM native. These layers are retired from the HCSM paper stack. None of them is cited by any HCSM native theorem. Each is either replaced by an HCSM-native theorem or is pending an HCSM native derivation, as stated explicitly below.

SYK/QIN substrate.

The Sachdev–Ye–Kitaev Hamiltonian on N Majorana fermions was used in the WIN-era stack to provide a microscopic model for chaotic scrambling. It was removed from the framework in Version 2 of the WIN Paradigm Explained: direct computation of the infrared structure of the Ω^1 fermionic sector establishes that site-local D_4 -invariant quartic interactions do not flow to SYK. The correct infrared description of Ω^1 is a $(2 + 1)$ -dimensional local fermionic theory, not SYK. In HCSM, this layer is retired entirely.

ENTROPIX/MESA.

The master entropy selection framework was used in the WIN-era stack to select the unique configuration consistent with the master conditions M1 and M2. It is not referenced by any HCSM-native theorem, and the HCSM paper stack does not depend on it. Retired.

Kaluza–Klein graviton at 1.52 TeV.

The WIN Gravity Theorem contained a Kaluza–Klein interpretation of the first excited graviton level. This interpretation imported the assumption that the lattice is a compactified spatial dimension, which contradicts the framework’s central result $\dim D = 4$ that spacetime emerges from the substrate. It is retracted as a category error in HCSM-49. Any detection of a discrete graviton spectrum at accessible scales would falsify HCSM.

5D/6D/7D/8D tower.

Several WIN era papers (particularly those on black hole evaporation and the cosmological constant) used a tower of extra dimensions labeled 5D through 8D, with an “Archival Persistence Layer” at 8D. None of these layers is HCSM native. They are retired.

Ryu–Takayanagi strip integral.

The DME stability coefficient $\gamma = 0.702839\dots$ was derived in the WIN era stack from the finite part of the holographic entanglement entropy of a strip via the Ryu–Takayanagi formula. The strip integral $I(\alpha) = \int_1^\infty du [u^{2\alpha-2}/\sqrt{u^{2\alpha}-1} - u^{\alpha-2}]$ is a continuum object. An HCSM-native derivation of γ from a discrete computation on the torus is pending (HCSM-55). Until that derivation is complete, the DME layer is treated as pending import removal, and no HCSM theorem depends on γ .

EDF Ohmic bath.

The Core gradient flow $d\epsilon/d\lambda = -2\gamma\epsilon$ was derived in the WIN-era stack from an open-system model with an Ohmic hidden bath in the overdamped limit. The Ohmic bath is a continuum construction. An HCSM-native derivation of the gradient flow from discrete dynamics on the torus is pending (HCSM-56).

ENTROPIA-EFT.

The equation of state $w = -1 - (1 - \gamma)^2/(2\pi)$, the bulk viscosity $\zeta H/\rho$, and the Proca mass $m_s = (M_P/N) \exp(-5N/3)$ were derived in the WIN-era stack from a 2D mass-regulated bubble and an EDF running with $\gamma_f = 1$. These are continuum constructions. An HCSM-native derivation is pending (HCSM-57). Note: the algebraic form of the EFT coefficient $c = 5/9 = (d + 1)/(2d + 1)$ has been derived from the observer pair (HCSM-53), but the Feynman-diagram-style derivation from a discrete torus computation remains open.

10 The HCSM paper stack and its status

The HCSM paper stack consists of the following papers. Each paper is assigned a status:

Writable.

The paper's content is 100% HCSM-derived in its own domain. The paper can be written now.

Pending.

The paper's domain contains exactly one open item. The paper is held until the blocker closes.

Retired.

The paper's content is subsumed by an HCSM-native theorem or by a retired layer.

Table 3: The HCSM paper stack.

#	Title	Status
0	Foundations, Notation, and Mapping	Writable (this paper)
1	The 8×8 Torus Substrate from $d = 4$	Pending ($N = d \cdot 2^d$)
2	The Hodge Complex on Λ	Writable
3	Lorentzian Signature from the Hodge Star	Writable
4	The 14-Mode Multiplet at $\lambda = 4$	Writable
5	Orbit Decomposition and Inverse Participation	Writable
6	Sector-Coarse Correspondence	Writable
7	Hypercharge Quantization	Writable
8	Anomaly Cancellation as a Theorem	Writable
9	Gauge Topology and Orientation-Flip $\mathbb{Z}_2$	Writable
10	The Self-Canceling Vacuum Mechanism	Writable
11	The $5 + 9$ Split of the 14-Mode Multiplet	Writable
12	The Hodge–HCSM Interface: Entropic Flatness	Writable
13	The Processor T : F/B -Class Correspondence	Writable
14	Winding Generator Spectrum and Class Partition	Writable
15	Gauge Symmetry of the Loop Operator and DNLS	Writable

#	Title	Status
16	Loop Operator on the Full 56-Dimensional Hodge Sector	Writable
17	Loop Operator Cartan Rotation	Writable
18	The Dynamical Bridge: DNLS on the Torus	Writable
19	DNLS Hartree Mass-Map Coefficient	Pending $((d - 1)/2)$
20	Motion Theorem	Pending (DME import)
21	Time Theorem	Pending (loop pass duration)
22	The Zero State	Writable
23	The δ Function and $N/4$ Labels	Writable
24	The Charge Theorem	Writable
25	The 4×4 Periodic Charge Density	Writable
26	The Observer Pair $\{W, t\}$	Writable
27	Klein Orbit Structure	Writable
28	Orbit-Class Reconstruction	Writable
29	Charge–Entropy–Observer Triangle	Writable
30	The Base-2 Mass Formula Pattern	Writable
31	Generation Count and Flavor Mechanics	Pending (generation count)
32	Shell Assignment Rule	Writable
33	Mass Map	Pending $((d - 1)/2)$
34	CKM Matrix	Pending (residual structure)
35	V_{td} Structural Relations	Pending (non-perturbative)
36	PMNS Matrix	Pending (0.23% overshoot)
37	PMNS Generation Phase	Pending
38	Hidden Sector Content	Writable
39	Weinberg Angle	Writable
40	Higgs Mechanism	Pending (Higgs origin)
41	Dark Photon and Kinetic Mixing	Writable
42	Spin and Chirality from the $\mathbb{Z}_2$ Grading	Pending (spin structure)
43	Strong CP from Orientation $\mathbb{Z}_2$	Pending
44	The Proca Sector	Pending ($U(1)$ origin)
45	Graviton as Self-Paired Curvature Mode	Writable
46	Graviton Exact Masslessness	Writable
47	Graviton Self-Coupling from D_4	Writable
48	Faddeev–Popov G_N Normalization	Writable
49	Newton Constant 8π Factor	Writable
50	Composite-Operator Effective Action	Writable

#	Title	Status
51	kL Crossover (Bare)	Writable
52	kL Crossover (Corrected)	Pending (DME import)
53	EFT Coefficient in ρ_{DE}	Writable
54	Dark Energy Density	Pending (needs HCSM-52)
55	Vacuum Stability (DME)	Pending import removal
56	Core Dynamics	Pending import removal
57	Vacuum EFT	Pending import removal
58	H_0 Boost	Pending
59	Joint Independence of the Geometric Kernel	Pending
60	Complete Numerical Verification Suite	Writable

Summary: 34 papers are writable now (100% HCSM-derived), 25 are pending one open item each, and 2 (SYK, ENTROPIX/MESA) are retired. The four critical blockers are: (1) $N = d \cdot 2^d$, (2) γ from HCSM-native strip computation, (3) loop pass duration, (4) $(d - 1)/2$ from higher-order DNLS. Resolving blocker 1 alone converts 6 pending papers to writable.

11 Reproducibility

All numerical claims in the HCSM paper stack are verified at machine precision in IEEE-754 double precision. The verification scripts are provided alongside each HCSM paper in the companion repository. Each script uses only standard libraries (NumPy, SciPy, SymPy, mpmath) and constructs the objects it needs from HCSM-native definitions.

The verification repository structure:

```
HCSM/
  00-FOUNDATIONS/
  01-SUBSTRATE/
  02-HODGE_COMPLEX/
  03-SPECTRAL_MIDPOINT/
  04-GAUGE_STRUCTURE/
  05-PROCESSOR/
  06-PROCESSOR_DYNAMICS/
  07-OUTPUT_STRUCTURES/
  08-FLAVOR_SECTOR/
  09-GAUGE_CONTENT/
  10-GRAVITY/
  11-COSMOLOGY/
  12-VERIFICATION/
  _MAPPING/
  _REFERENCES/
  _LOGS/
  _FIGURES/
  _TEMPLATES/
```

Each numbered folder corresponds to an HCSM paper. Each paper’s verification script is placed in 12-VERIFICATION/ with the same number. Running 12-VERIFICATION/run_all.py reproduces every numerical claim in the stack.

12 Conclusion

This paper has established the foundation of the Hodge Complex Standard Model. The substrate is the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, selected by two independent anchors (photon helicity for $d = 4$, equal entropy spacing for $L = 8$) and a derived third anchor (the topology of the Hodge–Dirac kernel). The substrate carries the Hodge complex $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$ with $D^2 = \Delta$ exactly on each form degree. The 14 modes at the self-paired eigenvalue $\lambda = 4$ are identified with one Standard Model generation. The processor T and the class partition are the machine’s operation. The output structures—charge, mass, flavor, gravity, cosmology—are the residue after mirror cancellation.

Every theorem cited in this document is HCSM-native. Where prior papers used imported machinery, this document states the verdict clearly. Retired layers are retired; pending layers are named with their specific blocker; writable layers are 100% HCSM-derived.

The complete HCSM paper stack consists of 60 numbered papers. At the time of this writing, 34 are writable, 25 are pending one open item each, and 2 are retired. The four critical blockers that unblock the entire remaining stack are: $N = d \cdot 2^d$, γ from HCSM-native computation, loop pass duration selection, and $(d - 1)/2$ from higher-order DNLS.

The framework’s claim is that the Standard Model is what the substrate looks like when you observe it. The Hodge complex is the machine. Physics is what the machine cannot cancel.

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A Web Crawler Metadata

A.1 Dublin Core

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 DC.creator: Preschutti, Stanley
 DC.creator.orcid: 0009-0004-5445-1744
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 DC.type: Text / Preprint / White Paper
 DC.format: application/pdf; text/x-latex
 DC.language: en
 DC.publisher: Entropia Research Institute
 DC.rights: All rights reserved
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A.2 Schema.org JSON-LD

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A.3 BibTeX for Automated Indexing

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  journal = {HCSM White Paper Series},
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  month = {September},
  note = {ORCID: 0009-0004-5445-1744},
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    Standard Model derivation, discrete substrate, foundations},
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}

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